

## Shay Mordechai

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Core Skills: **Web PT** • **TTPs** • **IDA Pro** • **Python** • **Network Analysis** • **Mobile API** • **Vulnerability Research**

## Summary

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Offensive Security Researcher specializing in vulnerability discovery, Reverse Engineering, and OS Internals. Proven expertise in bypassing anti-analysis mechanisms, auditing complex execution environments (JIT, Compilers), and authoring comprehensive and technically rigorous reports.

## Professional Experience

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### Security Architect | MyLeads AI | Jan 2026 – Present

- **Red Team:** Executed Web PT via Burp Suite, developing advanced TTPs to uncover undisclosed APIs, logic flaws, and IDORs.
- **Secure Architecture & Isolation:** Architected Envoy Proxy tunnels to prevent data leakage, and hardened OS execution via Rootless Podman & Linux namespaces.

### Application Security Intern | Israel Tax Authority | Aug 2024 – Jul 2025

- **AppSec:** Integrated Checkmarx SAST/SCA into 15+ CI/CD pipelines, trained R&D on OWASP Top 10, and hardened IBM DataPower gateways via strict JSON validation schemas.

**Combat Soldier | Intelligence Collection Corps, IDF | Aug 2016 – Apr 2018** - Executed covert reconnaissance operations, deploying stealth observation posts for operational target profiling.

## Education

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**B.Sc. Computer Science | Jerusalem College of Technology | 2021 – 2025 (GPA: 89)** - **Coursework:** Reverse Engineering (93), Network Analysis (97), InfoSec (90).

## Projects & Research (*Detailed Write-ups at [shay-mordechai.github.io](https://shay-mordechai.github.io)*)

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- **Mobile API Security:** Intercepting Android traffic via Frida and Objection to bypass SSL Pinning/Root Detection. Static analysis of APKs using JADX.
- **Advanced RE Challenges:** Solved highly selective challenges (ISA/Shabak Level 1). Analyzed multi-stage malware, bypassed TLS Callbacks, and extracted PE payloads.
- **AI-Augmented Vuln Research:** Executed pre-deployment red-teaming via semantic fuzzing, discovering 3 pending Zero-Days (CERT-IL #11265).
- **Network Exfiltration & C2:** Engineered a CTF platform modeling nation-state tactics, implementing ICMP covert channels, PKI MITM, and steganography.
- **V8 JIT Internals:** Published comprehensive research on browser engine vulnerabilities, focusing on RWX memory allocation patterns.
- **TLS 1.2 Simulator:** Authored a 4,000-line Python/Scapy framework simulating full TLS 1.2 handshakes from scratch, including SSLKeyLog generation for DPI.

## Technical Skills

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- **Tools:** IDA Pro, JADX, x32dbg, WinDbg, Wireshark, Burp Suite, Sysinternals, Checkmarx, Frida.
- **Languages:** C/C++, x86 Assembly, Python, Bash. Knowledge of JavaScript, Java, C#, Rust, Raku.
- **Bug Bounty & CTFs:** PSIRT Acknowledgments (Palo Alto Networks & TP-Link). Active in advanced RE environments (crackmes.one, Flare-On).